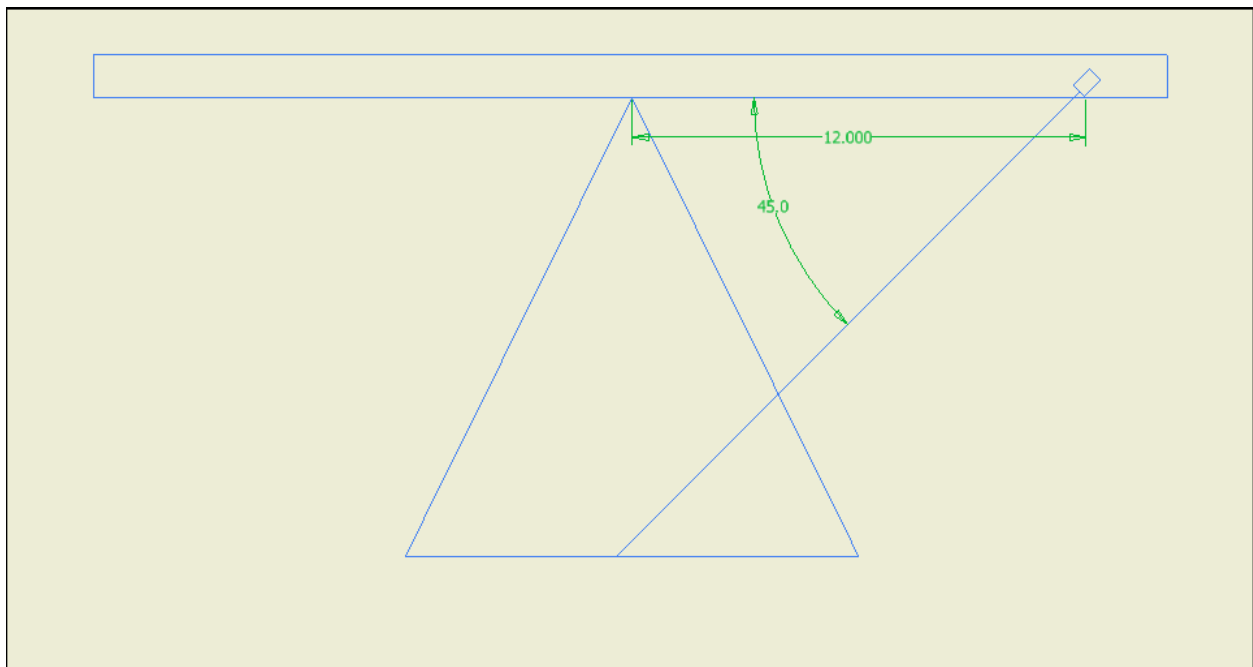


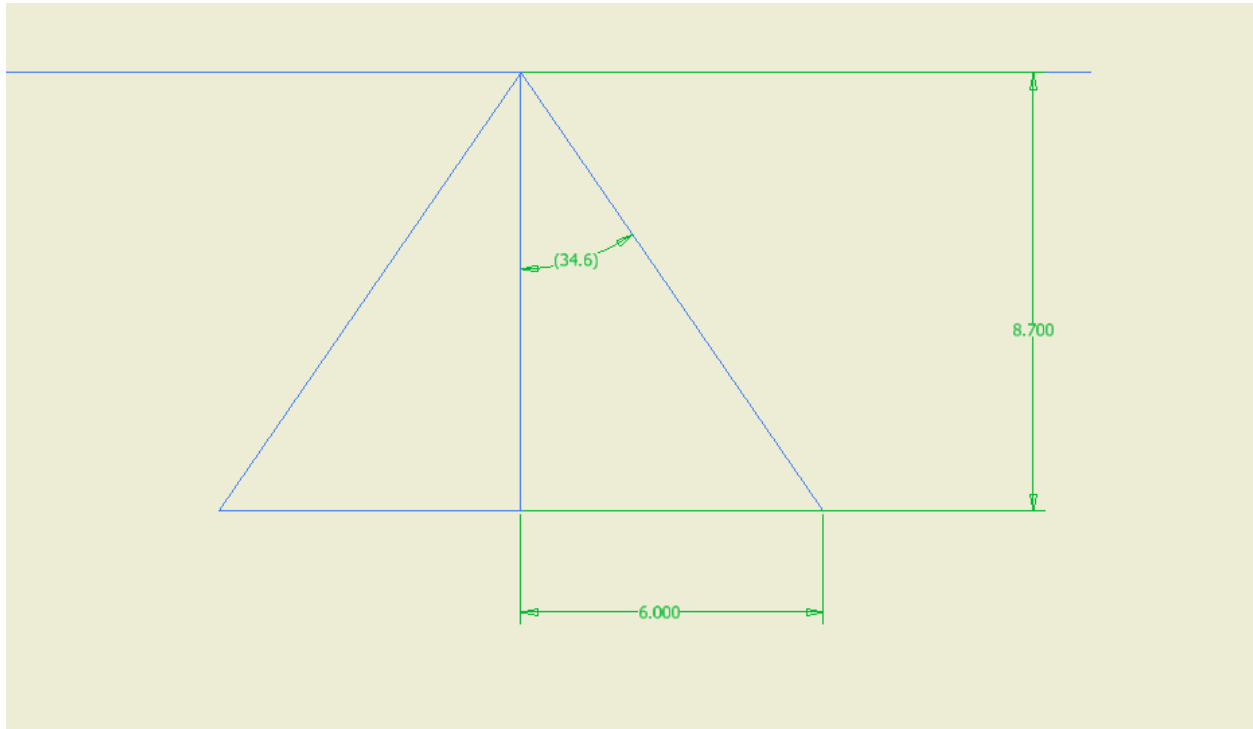
```

def Interpolate(x1, x2, y1, y2, x):
    if x2-x1==0:
        return -1
    m = (y2-y1)/(x2-x1)
    b = y1-m*x1
    y = m*x+b
    return y

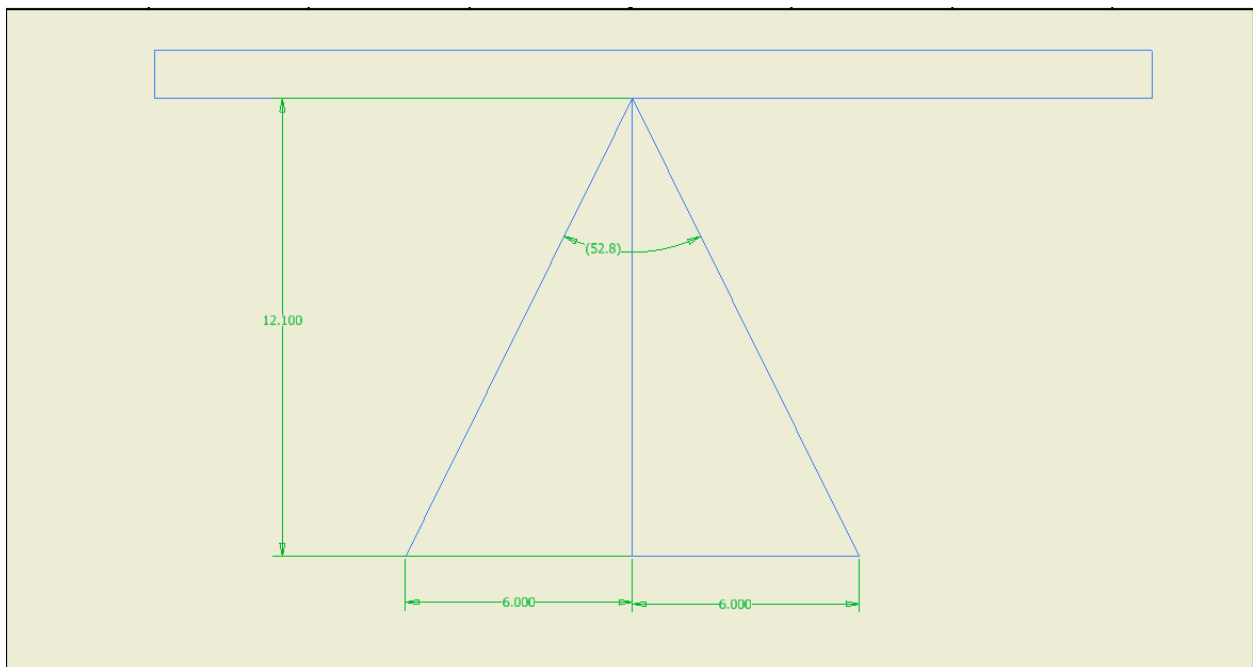
for guess in np.arange (0,50,0.01):
    xlimit = guess*math.tan(CameraFrustrumAngle/2*(math.pi/180))
    xIntercept = IntersectionOfLaserAndGuess(guess, DistanceFromCameraToLaser, AngleOfLaser)
    pixelNumber = int(Interpolate(-xlimit, xlimit, PIXELWIDTH, 0, xIntercept))

```





```
pixelNumber = int(Interpolate(-xlimit, xlimit, PIXELWIDTH, 0, xIntercept))
```



```
def IntersectionOfLaserAndGuess(GuessedDistance, DistanceFromCameraToLaser, AngleOfLaser):
    temp = math.tan(AngleOfLaser*math.pi/180.0)
    x = (GuessedDistance-DistanceFromCameraToLaser*temp)/temp
    return x
```

```
xlimit = guess*math.tan(CameraFrustrumAngle/2*(math.pi/180))
```

